

A $c_0(\Gamma)$ Subcase of the WLD Projective Tensor Question

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2026-06-18

Status

This is a candidate partial result for Question 8(b) of Aviles, Martinez-Cervantes, Rodriguez, and Rueda Zoca, *Topological properties in tensor products of Banach spaces*, arXiv:2202.00371.

Source Question

Question 8(b) asks:

Suppose that X and Y are WLD and that either X or Y has the Dunford-Pettis property.
Is $X \widehat{\otimes}_\pi Y$ WLD?

The source paper proves the characterization

$$X \widehat{\otimes}_\pi Y \text{ is WLD}$$

if and only if X and Y are WLD and every operator

$$X \rightarrow Y^*, \quad Y \rightarrow X^*$$

has norm-separable range.

Result

Theorem 1. *Let Γ be any index set and let Y be a WLD Banach space. Then*

$$c_0(\Gamma) \widehat{\otimes}_\pi Y$$

is WLD. In particular, Question 8(b) has a positive answer whenever one factor is $c_0(\Gamma)$.

Proof Intuition

The source characterization reduces the tensor question to two separable-range operator checks. The hard-looking direction is controlled by a simple WLD obstruction: a WLD space cannot have a nonseparable image inside any $\ell_1(\Gamma)$. Otherwise the image closure would be a nonseparable WLD subspace of $\ell_1(\Gamma)$, hence would contain $\ell_1(\omega_1)$, but $\ell_1(\omega_1)$ fails Corson's property (C), while WLD spaces have property (C) hereditarily.

For operators $c_0(\Gamma) \rightarrow Y^*$, adjoints first give separability of the coordinate functionals in $\ell_1(\Gamma)$. This makes the weak-star topology on the image ball countably determined; the source paper's WLD lemma then upgrades weak-star separability of the image to norm separability.

Proof

Lemma 1. *Every nonseparable subspace of $\ell_1(\Gamma)$ contains an isomorphic copy of $\ell_1(\omega_1)$.*

Proof. This is standard, but we recall the argument. Let $Z \subseteq \ell_1(\Gamma)$ be nonseparable. There are $\varepsilon > 0$ and an uncountable family $(z_\alpha) \subset B_Z$ with $\|z_\alpha - z_\beta\| > \varepsilon$ for $\alpha \neq \beta$. Approximate each z_α within $\varepsilon/16$ by a finitely supported vector p_α . By the finite Δ -system lemma and a further thinning, the supports of the p_α 's have a common finite root R , the root parts $p_\alpha|_R$ are pairwise within $\varepsilon/16$, and the off-root supports are pairwise disjoint.

Let $q_\alpha = p_\alpha|_{\Gamma \setminus R}$. After discarding at most countably many terms and thinning again, $\|q_\alpha\| \geq \delta$ for some $\delta > 0$. Choose the original approximation error small enough compared with δ . For any finite scalar family (a_α) , projection onto $\Gamma \setminus R$ gives

$$\left\| \sum_{\alpha} a_{\alpha} z_{\alpha} \right\| \geq \left\| \sum_{\alpha} a_{\alpha} q_{\alpha} \right\| - \sum_{\alpha} |a_{\alpha}| \|z_{\alpha} - p_{\alpha}\| \geq c \sum_{\alpha} |a_{\alpha}|$$

for some $c > 0$, because the q_α 's have disjoint supports and norms bounded below. The reverse ℓ_1 -estimate is immediate from $\|z_\alpha\| \leq 1$. Thus an uncountable subfamily of the z_α 's is equivalent to the canonical basis of $\ell_1(\omega_1)$. $\square$

Lemma 2. *Let X be WLD and let $T : X \rightarrow \ell_1(\Gamma)$ be bounded. Then $T(X)$ is norm separable.*

Proof. Let $Z = \overline{T(X)}$. The operator $T : X \rightarrow Z$ has dense range, so $T^* : Z^* \rightarrow X^*$ is injective and weak-star to weak-star continuous. Since X is WLD, X^* admits an injective weak-star-to-pointwise map into some $\ell_\infty^c(I)$. Composing with T^* shows that Z is WLD.

If Z were nonseparable, Lemma 1 would give a subspace of Z isomorphic to $\ell_1(\omega_1)$. But WLD spaces are hereditary and have Corson's property (C), while $\ell_1(\omega_1)$ fails property (C) (as also used in Section 3 of the source paper). This contradiction proves that Z , and hence $T(X)$, is separable. $\square$

Proof of the theorem. The space $c_0(\Gamma)$ is WLD and has the Dunford-Pettis property. By the source paper's Theorem 4.1, it remains to prove that every operator in both cross directions

$$c_0(\Gamma) \rightarrow Y^*, \quad Y \rightarrow c_0(\Gamma)^* = \ell_1(\Gamma)$$

has separable range.

The second direction is exactly Lemma 2.

For the first direction, let $S : c_0(\Gamma) \rightarrow Y^*$ be bounded. Consider the adjoint restricted to the canonical copy of Y in Y^{**} :

$$S^*|_Y : Y \rightarrow c_0(\Gamma)^* = \ell_1(\Gamma).$$

By Lemma 2, the range $S^*(Y)$ is separable. Choose a countable set $(y_n) \subset Y$ such that S^*y_n is norm dense in $S^*(Y)$.

For $x \in B_{c_0(\Gamma)}$ and $y \in Y$,

$$Sx(y) = S^*y(x).$$

Since every S^*y is uniformly approximable on $B_{c_0(\Gamma)}$ by some S^*y_n , the weak-star topology on $S(B_{c_0(\Gamma)}) \subset Y^*$ is determined by the countable family (y_n) . Therefore the weak-star closure of $S(B_{c_0(\Gamma)})$ is compact metrizable, hence $S(B_{c_0(\Gamma)})$ is weak-star separable.

The source paper's Lemma 4.2 says that if E and F are WLD and $R : E \rightarrow F^*$ is bounded, then $R(E)$ is norm separable whenever it is weak-star separable. Applying this to $E = c_0(\Gamma)$, $F = Y$, and $R = S$, we conclude that S has norm-separable range.

Thus all cross operators have separable range, and the source characterization implies $c_0(\Gamma) \widehat{\otimes}_\pi Y$ is WLD. $\square$

Verification Notes

The proof uses only standard permanence facts for WLD spaces, the standard subspace structure of $\ell_1(\Gamma)$, and two explicit results from the source paper: Theorem 4.1, which characterizes WLD projective tensor products by separable cross-operator ranges, and Lemma 4.2, which upgrades weak-star separability to norm separability for operators $X \rightarrow Y^*$ when both spaces are WLD.

Limitations and Novelty Check

This is a partial result. It does not solve Question 8(b) for arbitrary WLD spaces with the Dunford-Pettis property.

The run's existing attempt for arXiv:2202.00371 had identified this $\ell_1(\Gamma)$ -range lemma as the most promising next route, but did not verify it. Local index searches found no existing packet for this subcase. Web searches for the exact source question and nearby phrases found the source arXiv page but no later direct answer.

References

- [1] A. Aviles, G. Martinez-Cervantes, J. Rodriguez, A. Rueda Zoca, *Topological properties in tensor products of Banach spaces*, arXiv:2202.00371.
- [2] M. Fabian, P. Habala, P. Hajek, V. Montesinos, J. Pelant, V. Zizler, *Functional Analysis and Infinite-Dimensional Geometry*, Springer, 2001.

Human Review Recommendation

Review as a likely valid partial result. The proof point most worth checking is Lemma 1, the standard extraction of an $\ell_1(\omega_1)$ -sequence from a nonseparable subspace of $\ell_1(\Gamma)$.